

TaskManagerPRO

Bento Dashboard, Kanban Board & Secure SQLite Architecture

PBKDF2 Password Hashing, Soft Delete, CSV Pipelines & Automated Backups

ENTERPRISE ARCHITECTURE & HIGHLIGHTS

- Bento Grid Dashboard: Interactive metric cards, status bars, and category breakdown
- Kanban Workflow: To Do -> In Progress -> Review -> Completed lifecycle
- Enterprise Security: 100% Parameterized queries + PBKDF2 Password Hashing
- Soft Delete Architecture: Zero accidental data loss with Trash recovery
- Category Management & CSV Engine: UTF-8 BOM CSV import/export pipeline
- Automated Backup Service: Timed database snapshots preventing corruption

PRIMARY AUTHOR & LEAD ARCHITECT

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Subject: Advanced Web & Modern AI Application Engineering (Lab 3 & 4)

1. System Requirements & Bento Grid UX

TaskManagerPRO was developed for Lab 3 & 4 to solve the challenge of fragmented task tracking. It implements a modern Bento Grid layout combining summary metric cards, interactive filter pills, and a real-time Kanban board.

Bento Metric Cards

Real-time task counters: Total Tasks, Completed, In Progress, and High Priority.

Kanban Workflow

Status transitions with drag/click: To Do -> In Progress -> Review -> Completed.

Category Badges

Custom color tags for categorizing tasks across projects and urgency levels.

Filter & Search Bar

Instant responsive search by keyword, category, status, and due date range.

2. Security Protocols & Cryptographic Authentication

PBKDF2 Password Hashing (auth_manager.py):

Never store plain-text passwords. TaskManagerPRO uses `hashlib.pbkdf2_hmac` with SHA-256 and 100,000 iterations combined with a cryptographically secure 16-byte random salt per user.

100% Parameterized SQLite Queries:

Every database query uses parameterized placeholder queries (`cursor.execute('SELECT * FROM tasks WHERE user_id = ?', (user_id,))`), providing total immunity against SQL Injection attacks.

3. Soft Delete Architecture & Zero Data Loss

Two-Stage Deletion Model:

When a user deletes a task, it is NOT permanently destroyed. Instead, the task is marked with `is_deleted = 1` and `deleted_at = CURRENT_TIMESTAMP`. The item is safely hidden from the dashboard and moved to the Trash View (`trash_view.py`) where users can restore it at any time or permanently purge it.

Automated Database Backup Service (`backup_service.py`):

A background service automatically creates rotating compressed SQLite backups on exit and startup, guaranteeing that power interruptions or OS crashes will never result in corrupted data.

4. CSV Import/Export & Testing Matrix

CSV Service with UTF-8 BOM Support (csv_service.py):

Exports tasks to CSV using utf-8-sig (BOM) encoding so Microsoft Excel on Windows renders Thai characters perfectly without corrupted symbols. Supports round-trip CSV import with field validation.

Verification Suite:

Authentication	PBKDF2 verification with invalid & valid passwords	PASS [x]
SQL Injection	Inputs containing ' OR '1'='1 treated strictly as literals	PASS [x]
Soft Delete	Task removed from main table, verified in TrashView	PASS [x]
Trash Restore	Task returned to active status with original metadata	PASS [x]
CSV Export	UTF-8 BOM file verified in Excel without font encoding bugs	PASS [x]

5. Production Packaging & Standalone .exe

TaskManagerPRO compiles into a standalone Windows executable via TaskManagerPRO.spec, bundling the SQLite database engine, icons, and themes into a self-contained production binary.

ACADEMIC VERIFICATION STAMP & SUBMISSION DETAILS

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Lab 3 & 4: TaskManagerPRO - Grade Verification: Submitted & Verified